Liyan Tan

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Zeroth-order and forward-only optimization for LLM fine-tuning, on-device training, and analog/RF circuit design.

Education

University of California, Santa Barbara, *Ph.D. in Computer Engineering* | CA, USA — Spring 2025 - Expected 2028
University of California, Santa Barbara, *M.S. in Computer Engineering* | CA, USA — Fall 2023 - Winter 2025
University of California, Santa Barbara, *Extension Program in ECE* | CA, USA — Fall 2022 - Spring 2023
Huazhong University of Science & Technology, *B.S. in Electronic Information Engineering* | China — Fall 2019 - Summer 2023

Industrial Experience

Cadence Design Systems, *Software Architect Intern* | Austin, TX — Jun 2025 - Sep 2025

- Architected an LLM copilot agent for Voltus, Cadence's power-integrity signoff solver, turning natural-language design intent into verified command invocations, GUI actions, and automated root-cause analysis over signoff reports.
- Enabled on-premise deployment under enterprise data constraints through retrieval over EDA documentation, parameter-efficient fine-tuning, and distillation of a large teacher into a compact local model.

Selected Publications

GRZO: Group-Relative Zeroth-Order Optimization for Large Language Model Fine-Tuning [paper]
Liyan Tan, Yequan Zhao, Yifan Yang, Ruijie Zhang, Xinling Yu, Zheng Zhang, *Findings of the Association for Computational Linguistics: EMNLP*, 2026.

Training Specialist LLMs on Your Phone: Forward-Only Optimization via Block-Spectral Coordinates [paper]
Liyan Tan*, Yequan Zhao*, Ziyue Liu, Yupeng Su, Ruijie Zhang, Zheng Zhang, *arXiv preprint arXiv:2611.04797*, 2026.
(*equal contribution)

ZOAF: Towards Efficient Zeroth-Order Optimization for Analog/RF Circuit Design [paper]
Liyan Tan, Yequan Zhao, Jinming Lu, Ben F. Jamroz, Ari Feldman, Zheng Zhang, under review at *IEEE TCAD*, arXiv:2606.02869, 2026.

Simulation-Efficient Analog Circuit Yield Optimization via Monte Carlo Zeroth-Order Gradient Estimation [paper]
Liyan Tan, Yequan Zhao, Ben F. Jamroz, Ari Feldman, Zheng Zhang, under review, arXiv:2609.30678, 2026.

FuRA: Full-Rank Parameter-Efficient Fine-Tuning with Spectral Preconditioning [paper]
Yequan Zhao, Ruijie Zhang, **Liyan Tan**, Niall Moran, Tong Qin, Zheng Zhang, *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

IAPO: Input Attribution-Aware Policy Optimization for Tool Use in Small Multimodal Agents [paper]
Yifan Yang, Zhen Zhang, Jiayi Tian, **Liyan Tan**, Zheng Zhang, *arXiv preprint arXiv:2606.11652*, 2026.

MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training [paper]
Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Yupeng Su, **Liyan Tan**, Zheng Zhang, *arXiv preprint arXiv:2602.21545*, 2026.

Overview and Idea of the Mars UAV Experimental Platform Development [paper]
Liyan Tan, Yijun Mao, *Advances in Aeronautical Science and Engineering*, 15(2):1-10, 2024.

Research Projects

Zeroth-Order and Forward-Only Optimization for LLMs | Advisor: Zheng Zhang — Mar 2025 - Present

- Proposed GRZO, a group-relative zeroth-order optimizer that assigns a pseudo-independent perturbation to every mini-batch example, raising the effective gradient-direction count from one to the batch size at no additional forward cost.
- Proved directional unbiasedness with variance shrinking proportionally to the batch size; at the same inference-level memory footprint and comparable wall-clock cost, improved Llama3-8B average accuracy by $+3.0$ over MeZO and lifted sparse, low-rank, and quantized ZO variants by $+4.9$ on average.
- Co-led MobileSpec, which trains task specialists of an on-device INT4 LLM with forward passes alone. Its search space, FuRA-S, rescales only the block singular values of pretrained weights (0.015–0.05% of parameters), which rotates their singular vectors the way RLVR does.

- With 1–5 MB specialists, lifted a phone-use agent from 6–23% to 89–100% success on three of five tasks, and trained a personalized 8B model on an iPhone 16 Pro overnight at 3.5 GB peak memory.

Zeroth-Order Optimization for Analog/RF Circuit Design | Advisor: Zheng Zhang, with NIST Jan 2025 - Present

- Proposed ZOAF, a surrogate-free framework recovering gradient-descent directions from a small number of black-box circuit simulations, with a hybrid schedule from random-direction exploration to coordinate-wise refinement, quasi-random multi-start, and a sliding-window monitor for early stopping and feasibility.
- Achieved the best median figure of merit on all three benchmark schematics, with up to an order-of-magnitude advantage in median peaking on a 22-parameter amplifier and 1.3-$3.8\times$ fewer simulator calls to convergence.
- Extended the approach to yield optimization under process variation with ZO-MC-SGD, which turns specification margins into Monte Carlo zeroth-order descent directions with no explicit dependence on process dimension; reached a mean yield of 0.95 on four of five benchmarks within 50–200 SPICE simulations, up to $8\times$ fewer than the best baseline.
- Next: distributionally robust circuit optimization that accounts for distribution shift in process variations.

Tensor-Train Compressed NN via Temporal Logic | Advisors: Zheng Zhang, Tim Sherwood Apr 2024 - Jan 2025

- Implemented tensor-train compressed neural networks on temporal-logic hardware using delay-space values and approximated arithmetic, reducing parameter count by $115\times$ at 2.21% (exact) and 4.72% (approximated) accuracy loss.
- Cut latency by $14.28\times$ and energy by $32.37\times$ on a 4×4 PE array under race-logic hardware settings.

Mars UAV Aerodynamic Performance Experimental Platform | Advisor: Yijun Mao, HUST Jan 2021 - Sep 2022

- Surveyed two decades of Mars UAV experimental platforms and summarized their design trade-offs; the resulting review was published in *Advances in Aeronautical Science and Engineering*.
- Built a hover-efficiency test platform with an axially distributed model, measuring rotational speed, torque, lift, and induced velocity of the rotor blades.

Teaching

Teaching Assistant, UC Santa Barbara: PHYS 127BL Digital Electronics; PHYS 127AL Analog Electronics; PHYS 6BL Introductory Physics; ECE 15A Fundamentals of Logic Design; PSTAT 5A Understanding Data; ECE 3 Introduction to Python Programming for Engineers.

Honors

Honours Degree for Undergraduates, Huazhong University of Science and Technology Jun 2023